School of Advanced Sciences
Department of Mathematics
Fall Semester 2025-26
BMAT201L Complex Variables and Linear Algebra

Tutorial -1

****

Answer at least two sub-questions from each of the following questions

1. Discuss the nature of the singularity for function

   (i) $f(z) = (z+1)\sin(\frac{1}{z-2})$

   (ii) $f(z) = \frac{z^2-1}{z^2(z+2)(z-1)}$

   (iii) $f(z) = \frac{\cot z}{z}$

2. Use Cauchy's integral formula to evaluate the following integral around a circle γ:

   (i) $\oint_\gamma \frac{\sin z}{(z-\pi/3)^3} dz;\ \gamma : |z| = 1$

   (ii) $\oint_\gamma \frac{\cos z}{z} dz;\ \gamma : |z| = 1$

   (iii) $\oint_\gamma \frac{z^2-z+1}{z-1} dz;\ \gamma : |z| = 3/2$

   iv $\oint_c \frac{5z+7}{z^2+2z-3} dz$ where C is the circle $|z-2| = 2$.

3. Compute the following contour integrals by using Cauchy's residue theorem:

   (i) $\oint_\gamma \frac{3z^3+z}{(z-1)(z^2+9)} dz;\ \gamma : |z-2| = 2$

   (ii) $\oint_\gamma \frac{z-1}{(z+1)^2(z-2)} dz;\ \gamma : |z-i| = 3$

   (iii) $\oint_\gamma \frac{ze^{2z}}{(z+1)(z+2)} dz;\ \gamma : |z| = \pi$

   (iv) $\oint_c \frac{1}{(z-3)^2(z-3)} dz$, where (i) C be the rectangular contour bounded by $x = 0, x = 4, y = -1, y = 1$ (ii) C be the circular contour $|z| = 2$.

   (v) $\oint_c \frac{2z+6}{z^2+4} dz$, where the contour C be the circle $|z-i| = 2$.

***

School of Advanced Sciences
Department of Mathematics
Fall Semester 2025-26
BMAT201L Complex Variables and Linear Algebra

Tutorial -2

****

Answer any THREE from

1. Evaluate $\oint_c \frac{z+1}{z^2(z-2i)}dz$, where the contour C be the circle (i) $|z| = 1$, (ii) $|z-2i| = 4$.

2. Evaluate $\oint_c \frac{z}{(z+1)(z^2+1)}dz$, where (i) $C : 16x^2 + y^2 = 4$ (ii) $|z| = 2$.

3. Evaluate using contour integration:

   (i) $\int_0^{2\pi} \frac{1}{(2+3\cos\theta)^2}d\theta$

   (ii) $\int_0^{\infty} \frac{1-\cos(x)}{x^2}dx$

4. Compute the Cauchy's principal value for the following integrals:

   (i) $\int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)^3}dx$

   (ii) $\int_{-\infty}^{\infty} \frac{x^2+x+2}{(x^4+10x^2+9)}dx$

   (iii) $\int_{-\infty}^{\infty} \frac{\cos x}{(x^2-10x^2+9)}dx$

***

School of Advanced Sciences
Department of Mathematics
Fall Semester 2025-26
BMAT201L Complex Variables and Linear Algebra

Tutorial - 3

****

Answer the following

1. Let $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$
   - Find A^{-1} using Cayley-Hamilton theorem
   - Compute A^4 using Cayley-Hamilton theorem
   - Determine eigenvalues and eigenvectors for A
2. Let $A == \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & -1 \end{bmatrix}$
   - What are the eigenvalues of A^{-1} and the matrix $B = A^3 - 2A2 + 3I$
   - Find the eigenvector that corresponds to the largest eigen value of A.
3. Find the values of b and c for which the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & b \\ 0 & b & c \end{bmatrix}$ has $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$ one eigenvector. For these values of b and c, calculate all eigenvalues and eigenvectors for A.

School of Advanced Sciences
Department of Mathematics
Fall Semester 2025-26
BMAT201L Complex Variables and Linear Algebra

Tutorial -4

****

Answer the following

1. Express the system $2x + y + z = 4; x + y - 2z = 3; -x - 2y + z = 1$ in the matrix form $AX = B$. Solve the system $AX = B$ using (i) Gauss elimination method (ii) Gauss-Jordon method. Hence, find a non-zero vector X such that $AX = \lambda X$.

2. Find the values of 'a' for which the following linear system has (i) no solution (ii) unique solution (iii) infinite solutions
$4x + y + z = 4; x + 4y - 2z = 4; 3x + 2y + (a^2 + 1)z = a + 6.$

3. Find an equation relating a, b and c, so that the linear system
$2x + 2y3z = a; 3x - y + 5z = b; x - 3y + 2z = c$
is consistent for any values of a, b and c. Find all solutions when that condition holds.

4. For some electrical circuit system, the flow of currents I_1, I_2 and I_3 is given by the linear system $I_1 + I_2 - I_3 = 0; 4I_1 + I_3 = 8; 4I_2 + I_3 = 16$. Find the currents I_1, I_2, I_3.

School of Advanced Sciences
Department of Mathematics
Fall Semester 2025-26
BMAT201L Complex Variables and Linear Algebra
Tutorial -5

****

Answer any TWO of the following

1. Let $W = \{(x, y, z, u) \in R^4 | x = 2y; z = 3u - y\}$ be subset the standard vector space R^4. Check whether W forms a subspace to R^4?
2. Let $S = \{(1, 1, 0), (1, 0, 2), (1, 1, 1)\}$ be a subset of R^3. Is the vector $v = (1, 2, -1)$ can be expressed as a linear combination of vectors in S?
3. Let V be the vector space of polynomials of degree at most 2. Let $v_1 = t^2 + 2t + 1, v_2 = t^2 + 2$ be in V. Does the set $\{v_1, v_2\}$ span V?
4. Find the basis for the subspace W of R^3 spanned by the vectors $\{(1, 2, 2), (3, 2, 1), (11, 10, 7), (7,$ what is the dimension of W?
5. Let $S = \{(1, 1, 0, 2), (1, 0, 2, -1), (1, 1, 1, 3)\}$ be a subset of R^4. Is the set S linearly independent?
6. Check whether the set $S = \{(1, 1, 1, 1), (1, 2, 3, 2), (2, 5, 6, 4), (2, 6, 8, 5)\}$ forms a basis for R^4.If not, find the dimension of the subspace they can span.

Examples:

1. $W = \{(x, y, z) \in R^3 : ax + by + cz = 0\}$, where a,b and c are scalars. Verify whether W is a subspace to V?
2. Which of the following subsets of R^2 are with usual addition and scalar multiplication are subspaces?

   a $W_1 = \{(x, y) : x \geq 0\}$

   b $W_2 = \{(x, y) : x, y \geq 0\}$

   c $W_3 = \{(x, y) : x = 0\}$

   d $W_4 = \{(x, y) : x + y = 1\}$

   e $W_5 = \{(x, y) : x - 2y = 0\}$
3. Let $V = E^3$ and $W = \{\mathbf{x} | \mathbf{x} = a(1, 0, 2) + b(1, -1, 3)\}$, where a and b are real numbers. Is W a subspace of V?
4. Let $V = M_{22}$ and $W = \{invertible\ 2 \times 2\ matrices\}$. Check whether W is a subspace to V?
5. Let $U = \{(a + 2b, 2a - 3b) | a, b \in R\}$. Is U a subspace of R^2? Why or why not?

6. Let V be the vector space P_2. Does $p = t^2 - 2t + 2$ can be expressed as in linear combination of $p_1 = t^2 + 2t + 1$, $p_2 = t^2 + 2$ and $p_3 = t - 1$.

7. Consider the vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

   - Show that the vectors $\mathbf{u}$ and $\mathbf{v}$ span $V = R^2$.
   - Write the vector $\mathbf{w} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

8. Let P_2 be the vector space of polynomials of degree at most 2. Let $p_1 = 2t^2 + t + 2$, $p_2 = t^2 - 2t$, $p_3 = 5t^2 - 5t + 2$ and $p_4 = -t^2 - 3t - 2$ are in P_2. Determine whether the vector $p = t^2 + t + 2$ belongs to span of $\{p_1, p_2, p_3, p_4\}$?

9. Let $V = R^3$. Check whether the vectors $(1, 2, 1), (1, 0, 2), (1, 1, 0)$ span V?

10. Determine which of the following sets are linearly independent and dependent:
(i) $S = \{(3, 1, 2), (3, 8, -5), (-3, 6, -9)\}$ in R^3.
(ii) $S = \{(2, 1, 3), (3, -1, 2), (10, 0, 10)\}$ in R^3.
(iii) $S = \{(1, 2, 1, -1), (4, 3, 1, 0), (2, 0, 1, 3)\}$ in R^4.

11. Let $\mathbf{x}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 4 \\ -7 \\ -1 \end{bmatrix}$ $\mathbf{x}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ belongs to the solution space of $A\mathbf{x} = \mathbf{0}$. Is the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ linearly independent?

12. Which of the following set(s) of polynomials span $V = P_2$:

    (a) $\{t^2 + 1, t^2 + t, t + 1\}$
    (b) $\{t^2 + 1, t - 1, t^2 + t\}$
    (c) $\{t^2 + 2, 2t^2 - t + 1, t + 2, t^2 + 2 + 4\}$

13. Are the vectors $[1, 0, 1, 2], [0, 1, 1, 2]$ and $[1, 1, 1, 3]$ in R^4 linearly dependent or independent?